Feasibility of EveNet for Low-Statistics NSBI in $HH \to b\bar{b}\tau^+\tau^-$ and $HH \to b\bar{b}\gamma\gamma$: Coverage Analysis, Tests, and Decision Plan

Working note — $b\bar{b}\tau^+\tau^-$ κ_λ NSBI project

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v1 — synthesised from arXiv:2601.17126, arXiv:2606.14870, and the local EveNet integration harness (branch `fm`)

Abstract

We assess whether the EveNet event-level foundation model (FM) [1] can be fine-tuned to train high-precision, compute-efficient density-ratio classifiers for a κ_λ measurement via neural simulation-based inference (NSBI) in $HH \to b\bar{b}\tau^+\tau^-$ and $HH \to b\bar{b}\gamma\gamma$, with emphasis on the low-statistics regimes: sparse signal Monte Carlo (MC) per coupling point (P1) and sparse systematic-variation MC (P2). The central finding: *neither hadronic taus nor prompt photons exist in EveNet's object vocabulary or pre-training corpus*, and no supported extension path is documented. However, a three-tier analysis (vocabulary / corpus / transferable structure) combined with an information-flow analysis of the NSBI ratio decomposition shows the gap is concentrated in identification information that approximately cancels in exactly the low-statistics ratio components where FM data-efficiency is needed. We therefore recommend proceeding with the already-built fine-tuning harness (Option A), upgraded with an "unoccupied-corner" type encoding and calibration-closure metrics, before considering vocabulary surgery (Option B) or a new τ/γ-aware FM (Option C). A staged decision plan with explicit gates and kill criteria is given, together with a ranked list of open questions. The single largest scientific unknown — whether fine-tuned FM classifiers remain calibrated enough for density-ratio use — is untested in the literature and is answerable at near-zero marginal cost within our sweep.

Contents

1 Scope and problem statement

Task. Measure the Higgs self-coupling modifier κ_λ via NSBI: train classifiers whose outputs convert to per-event density ratios $r(x;\kappa_\lambda,\kappa'_\lambda)$ between process hypotheses, combined into an unbinned likelihood. Signal discrimination between κ_λ hypotheses is driven by the m_{HH} lineshape and the triangle–box interference (non-monotone in κ_λ, with a near-degeneracy around $\kappa_\lambda \approx 2.45$), i.e. by event-level kinematics rather than any single-object property.

Channels. Primary: $HH \to b\bar{b}\tau^+\tau^-$ (2 b-jets, 2 taus — here focused on hadronic taus τ_h — plus missing transverse energy, MET, from neutrinos). Secondary increment: $HH \to b\bar{b}\gamma\gamma$ (2 b-jets, 2 prompt photons, no intrinsic MET from the photon leg).

The two low-statistics problems.

P1 Sparse signal MC per coupling basis point: $\mathcal{O}(200\text{k})$ events per $\kappa_\lambda \in \{0,1,5\}$, produced with Powheg $\to$ Pythia8 $\to$ Delphes $\Rightarrow$ high-variance ratio estimators.

P2 Sparse systematic-variation MC $\Rightarrow$ difficulty parameterising $s(x;\kappa_\lambda,\theta)$ in nuisance parameters θ; variation samples are typically 10×–100× smaller than nominal.

Quality gates already adopted by the analysis. Integral closure $< 1\%$ per basis process and shape closure at the 5% level (`CLOSURE_TOL` $= 0.05$, treated as fixed — a 5% bias is already significant). Any FM-based classifier must pass the *same* gates as the from-scratch baseline; area-under-curve (AUC)-type wins alone are insufficient for likelihood-ratio use.

Compute framing. NSBI multiplies training runs: (processes) × (systematic variations) × (ensemble members) × (κ_λ grid). Any FM must win on *value per compute* across this multiplicity, not on a single-classifier benchmark.

2 Evidence base

2.1 The Sophon null and the horizon/objective decomposition

Fine-tuning the single-jet tagger backbone Sophon [verify: Sophon arXiv ID; believed 2405.12972] [4] for $\kappa_\lambda = $ 0-vs-5 discrimination did *not* beat from-scratch training (scratch $\geq$ finetune $>$ frozen), and a cheap kinematic-ceiling BDT/MLP matched all three. Diagnosis (per `FOUNDATION_MODEL_BRIEF.md`): two independent mismatches,

(i) **horizon** — single-jet pretext vs. event-level physics (the κ_λ signal is a multi-object correlation), and

(ii) **objective** — discrete flavour classification vs. a continuous, interference-driven ratio.

EveNet is the cleanest available experiment for *disentangling* these: it is event-level but *not* coupling-aware. An EveNet null would imply the objective mismatch is binding; an EveNet win would implicate the horizon mismatch as the cause of the Sophon null. This disentangling logic is the primary scientific motivation for the EveNet test, independent of the τ/γ question.

2.2 Controlled pre-training-objective study (arXiv:2606.14870)

Elsharkawy et al. [2] compare pre-training objectives on a fixed backbone (OmniLearned Point–Edge Transformer, PET; 100k/7M/51M parameters) pre-trained on JetClass (100M jets; MadGraph5+Pythia8+Delphes 3.4.3) [3], fine-tuned on top tagging with $N_{\text{ft}} \in \{10^3, 10^4, 10^5, 1.2 \times 10^6\}$ and on JetNet generation. Findings (labels F1–F4 as in the companion FM/NSBI note):

F1 Classifier+MPM wins the low-label regime. At $N_{\text{ft}} = 10^4$ the classifier + masked-particle-modelling (MPM) hybrid is the top configuration at every model size (by their Data-Efficiency Score); with plentiful labels ($N_{\text{ft}} = 1.2\text{M}$) pure classifier pre-training is best. "Low-label" means few labelled downstream training events; in simulation-based HEP every event is labelled for free, so for us it simply means *low-statistics fine-tuning samples*. Our P2 samples (10^3–10^4) sit squarely in the winning regime; our 200k/κ_λ-point signal samples sit near the boundary where the advantage starts narrowing.

F2 Generative pre-training does not transfer to classification. Pure flow-matching pre-training matches or underperforms from-scratch at all sizes.

F3 Classification and generation are orthogonal. Flow matching must be in the pre-training mix to obtain downstream generative advantage, and vice versa.

F4 Pre-training validation loss is an imperfect downstream proxy (Pearson $r \approx -0.75$); downstream performance peaks *before* pre-training converges. Checkpoint selection is an open problem — for NSBI it becomes a *calibration-selection* problem.

Two additional results matter directly for the present decision:

- **Pre-training data volume is nearly irrelevant at low labels.** In their $10^5 \to 10^8$ pre-training-set scan (Micro model), scaling slopes with scarce downstream labels are essentially flat ($\bar{m} = 1.4$, vs. $\bar{m} = 182.5$ with abundant labels); only standalone MPM retains a positive slope. *Objective choice, not corpus size, drives the low-label gain.* This substantially reduces the corpus-production cost of any continued-pre-training option (Sec. 7).

- **Two evidence gaps.** (a) *Calibration is never tested*: all classification metrics are rejection-type ($1/\varepsilon_b$ at $\varepsilon_s = 0.3$). Whether a fine-tuned FM classifier converts to an unbiased density ratio is unknown. (b) *No genuinely out-of-domain transfer is tested*: JetClass→top-tagging is class-overlapping (tops are in the pre-training corpus); no downstream task involves object types absent from pre-training. Both gaps sit exactly on our critical path.

2.3 EveNet (arXiv:2601.17126): what it is

2.3.1 Input schema

Per-object token = 7 features: (E, p_T, η, ϕ, `btag`, `isLepton`, `charge`), with E, p_T log-normalised. Global (per-event) conditioning = 10 features: (MET, MET_ϕ, n_ℓ, n_b, n_j, H_T, H_T^ℓ, M_{all}, $M_{\ell\ell}$, M_{bb}). **MET is not a token** — it enters only through the global block. There is *no type-embedding vocabulary*: object identity is carried entirely by the three discrete flags. There is no `isTau`, no `isPhoton`, no `isMET`, and no extensible type table. The schema is fixed jointly by the `event_info` YAML used in preprocessing *and* training; changing it changes the input dimension of the `GlobalEmbedding`/input blocks and breaks strict checkpoint compatibility.

2.3.2 Pre-training corpus

$\sim$33 billion simulated events, $\sim$500M retained after preselection. Processes (their Table 1): QCD multijet (2–$4j$); $t\bar{t}$ (hadronic/1L/2L $+0, 1j$); $t\bar{t}W$, $t\bar{t}Z$; W+jets, Z+jets $(0, 1, 2j)$; dibosons WW, ZZ, WZ; and SM $H \to WW^*$ (0L/1L/2L). Chain: MadGraph5_aMC@NLO → Pythia → Delphes with a *generic* detector card (explicitly neither CMS nor ATLAS). "Leptons" throughout means electrons or muons. **No prompt-photon process (no $H \to \gamma\gamma$, γ+jets, $t\bar{t}\gamma$) and no hadronic-tau final state appears in the corpus.** Hadronic taus appear in the paper only as an event *veto* in the $X \to YH$ selection, never as input tokens.

2.3.3 Pre-training objectives

Two-stage, diffusion-time-conditioned:

- **Stage 1 (SSL only):** masked-particle reconstruction via diffusion inpainting (mask a subset of tokens; denoise conditioned on the rest), $\mathcal{L} = \|v - v_{\mathrm{pred}}\|_2^2$, masking probability ramped during training.

- **Stage 2 (multi-task):** classification (process discrimination, noise-weighted $\alpha^2(t)$ CE, decay-channel variations *absorbed within* each class), SSL generation, and *supervised generation = neutrino reconstruction* (denoise injected tokens for undetectable components conditioned on observed objects). Losses summed directly, no gradient surgery. Segmentation and assignment heads exist but are fine-tune-only.

No systematics-aware or coupling/nuisance-parameterised component exists in the pre-training objective.

2.3.4 Fine-tuning interface and demonstrated results

Three initialisations are compared throughout: *Scratch*, *SSL* (Stage-1 weights), *Full* (Stage-1+2). Standard recipe: joint fine-tune with backbone learning rate = 10% of head learning rate; optional adapters between transformer blocks (trainable parameters $\sim$20M→8M). Demonstrated tasks:

(a) $X \to YH \to b\bar{b}\,WW^* \to b\bar{b}\,\ell\nu qq'$ (CMS Open Data full-sim, 121 mass points, 1k–10k signal events each): Full > Scratch by ∼50% in significance-improvement characteristic (SIC); > XGBoost by ∼20%, > TabPFN by ∼10%; $\sim 3\times$ faster convergence.

(b) **Exotic** $H \to aa \to 4b$ ($m_a = 30\,\mathrm{GeV}$; explicitly absent from the corpus): SIC 4.1 (Full) vs. 1.6 (Scratch) vs. 1.4 (SPANet); at *5% of training data* SIC 1.7, exceeding both baselines at full data; pairing efficiency 58% vs. 26%.

(c) $t\bar{t}$ **dilepton spin correlations** (scan 1.5%–150% of 2.83M events): at 1.5%, $\Delta D = 1.85\%$ (Full) vs. 2.68% (Scratch); lepton–quark pairing 48% vs. 24%; extraction reported stable under JES-type shifts.

(d) $\Upsilon \to \mu\mu$ **rediscovery in real CMS Open Data** (weak supervision, ∼12k events): median significance $7.6 \pm 0.4\sigma$ (Full) vs. 1.45σ (Scratch) vs. 6.4σ published benchmark — the starkest low-statistics pre-training win.

Cautionary result: *mass-parameterised* fine-tuning showed "systematically reduced sensitivity" (their explanation: kinematic diversity across the mass grid limits parameter sharing). This warns against a κ_λ-parameterised head as the first NSBI design; our per-basis-point + linear-combination construction sidesteps it.

2.3.5 Architecture, compute, availability

PET encoder, 18.8M parameters (hidden 256, 16 layers, 16 heads, 10 local kNN layers); task heads 1.3–8.7M. Pre-trained on 512 A100s (Perlmutter), LION + EMA, PyTorch 2.6 + Lightning + Ray. Code is public at `github.com/UW-EPE-ML/EveNet_Public`; checkpoints are public at `huggingface.co/Avencast/EveNet` (`checkpoints.20M.a4.last.ckpt` and `SSL.20M.last.ckpt`); a `network-100M.yaml` scaled template exists in the repository. Scope restriction: resolved regime only (no boosted/large-R topologies). Robustness to the generic-Delphes→CMS full-sim domain shift is demonstrated *empirically*, not by construction.

3 The τ/γ coverage gap: three-tier analysis

"Does EveNet cover τ/γ final states?" decomposes into three separable questions with different remediation costs:
The trap to avoid is collapsing these into a binary "EveNet doesn't do taus ⇒ build our own FM." Tiers 1–2 are established facts with cheap or medium remedies; tier 3 is an empirical question our existing harness can answer directly.

4 Why the gap is likely non-binding: ratio-component information asymmetry

The NSBI likelihood decomposes into per-process density ratios, and the τ/γ *identification* information is asymmetrically distributed across them:

Signal–signal ratios $r(x; \kappa_\lambda, \kappa'_\lambda)$**.** Numerator and denominator share an *identical final state.* Type-identification quantities (DeepTau-like scores, decay modes, photon ID) are (to good approximation) common factors that cancel in the ratio. The κ_λ dependence lives in the m_{HH} lineshape and triangle–box interference — functions of the object four-vectors, which the stock token (E, p_T, η, ϕ) carries *exactly.*

Tier	Finding	Verdict	Remedy if binding
Vocabulary (can the schema represent τ_h/γ?)	No type table exists; identity = 3 flags (`btag`, `isLepton`, `charge`). No τ/γ flag. MET never a token.	Gap real, but shallower than feared (Sec. 5)	Corner encoding (free) or input-widening surgery (cheap–medium)
Corpus (did pre-training see τ/γ physics?)	Zero prompt-photon processes; zero τ_h final states; τ_h used only as an event veto.	Gap real	Continued pre-training on τ/γ-enriched corpus (medium; volume requirement reduced by the flat low-label scaling result of [2])
Transferable structure (does the learned representation help anyway?)	All demonstrated out-of-corpus successes ($H \to aa \to 4b$, $X \to YH$, fast-sim→full-sim, sim→data) are *new topologies of known object types*. The truly out-of-vocabulary case is untested anywhere in the literature.	**Open — the decisive question**	Not remediable by construction; must be *measured* (Gate G0, Sec. 8)

Table 1: Three-tier decomposition of the τ/γ coverage gap.

Signal-vs-background discriminators. Here τ-ID/γ-ID genuinely matter (fake rejection against $t\bar{t}$, QCD, γ+jets). But these components are *MC-rich* — $t\bar{t}$ is the most abundant sample in the analysis — so they are not where FM data-efficiency is needed, and existing full-feature classifiers can be retained for them.

Systematic-variation ratios (nominal/varied, P2). Variation samples share the nominal final state; type-ID again approximately cancels, and energy-scale-type shape effects are kinematic and token-visible.

Consequence. The ratio components where the FM's low-label advantage is needed (P1 signal ratios, P2 variation ratios) are precisely the components where the missing τ/γ vocabulary costs the least. The stock-schema workaround already encoded in the harness ($\tau_h \to$ generic jet) is approximately lossless *for the specific low-statistics question the FM is being asked to answer*, with the losses concentrated in MC-rich background rejection.

Honest caveats.

(i) "Fully preserved" applies to *visible* kinematics. In $b\bar{b}\tau^+\tau^-$, tau neutrinos smear visible m_{HH}; SVfit/FastMTT $m_{\tau\tau}$ — which partially recovers the neutrino momenta — has no slot in the stock schema and *is* genuinely lost. Mitigations: MET (the aggregate neutrino footprint) is in the global conditioning, and EveNet's Stage-2 pre-training explicitly included *neutrino reconstruction as a supervised denoising task* — the backbone was trained to infer invisible components from visible tokens + MET. Whether this recovers what FastMTT provides is an empirical question the sweep answers (open question Q5).

(ii) The cancellation argument is approximate: selection efficiencies and fake contamination differ slightly between κ_λ hypotheses through kinematic correlations with ID variables. Second-order, but the closure gates will price it.

(iii) For $b\bar{b}\,\gamma\gamma$ the preservation argument is *stronger* than for $b\bar{b}\,\tau^+\tau^-$: photons produce no neutrinos, so visible kinematics = full kinematics and $m_{\gamma\gamma}$ is exactly computable from the tokens. Conversely the FM *headroom* is smaller (the m_{HH} observable is already cleaner), which is why $b\bar{b}\,\gamma\gamma$ is sequenced as an increment, not the primary test.

5 Encoding strategies within the fixed schema

5.1 Status quo: anonymous-jet encoding

The current adapter (`scripts/delphes_to_evenet_npz.py`) emits τ_h as a generic jet: (`btag`, `isLepton`, `charge`) $= (0, 0, 0)$. Kinematics exact; τ identity, charge, and decay mode discarded; the network must infer τ-ness from kinematics alone. Documented, deliberate, checkpoint-safe.

5.2 Unoccupied-corner encoding (proposed, zero surgery)

The three flags span a $2 \times 2 \times 3 = 12$-point discrete lattice; every token's *type* is a lattice corner. In EveNet's pre-training corpus only four corners ever occur:

Corner (`btag`, `isLep`, `chg`)	**Corpus occupant**	**Proposed reassignment**	**Status**
$(0, 0, 0)$	light jet	—	occupied
$(1, 0, 0)$	b-jet	—	occupied
$(0, 1, \pm 1)$	$e^\pm/\mu^\pm$	—	occupied
$(0, 0, \pm 1)$	*never seen*	τ_h ("charged jet"), with measured charge	vacant
$(0, 1, 0)$	*never seen*	γ ("neutral lepton")	vacant
$(1, 1, \cdot)$, $(1, 0, \pm 1)$	*never seen*	reserved	vacant

Table 2: The flag lattice: occupied vs. vacant corners, and the proposed τ/γ assignments. Values are in-range for the pre-trained normalisers; the checkpoint loads unchanged.

This gives the network a *distinguishable* type signal for τ_h (recovering its measured charge, currently discarded) and γ inside the existing 7 dimensions — no weight surgery, full checkpoint compatibility, a one-line adapter change per object type.

5.3 Risks of the corner encoding (candidate failure mechanisms)

The backbone has no prior *about* vacant corners; "strictly more information at the data level" does not guarantee a better classifier after a *finite* fine-tune. Candidate mechanisms by which it could hurt, all measurable by an A/B test (Gate G1):

(R1) **Out-of-distribution embedding:** inputs at never-seen flag combinations land the token embedding in unexercised regions of representation space; early fine-tuning could be slower or less stable (mitigated by the backbone-LR= 10% recipe).

(R2) **Shortcut/overfit capacity at very low N_{ft}:** an extra clean discrete bit gives the network memorisation shortcuts; at $N_{\text{ft}} \sim 10^3$ this could *worsen* generalisation relative to the anonymous encoding.

(R3) **Attention misrouting:** pre-trained heuristics keyed on `charge` $\neq 0 \Rightarrow$ lepton may initially misroute τ_h tokens through lepton-like attention patterns (transient, but could matter in short fine-tunes).

5.4 What cannot be encoded without surgery

τ-ID score (DeepTau-like), decay mode, photon-ID quality, and FastMTT/SVfit $m_{\tau\tau}$ have no slot. Adding any of them widens the token (or the global block) and breaks direct checkpoint loading — that is Option B territory (Sec. 7). Overwriting one of the 10 global conditions with $m_{\tau\tau}$ is technically possible but abuses a pre-trained normaliser and is not recommended.

6 Current integration state (branch `fm`)

Already built and locally verified, in `examples/HH_bbtautau_kappalambda_sophon/`:

- **Adapter** `scripts/delphes_to_evenet_npz.py`: Delphes ROOT $\rightarrow$ NPZ with `x` of shape $(N, 16, 7)$ (zero-padded, p_T-sorted), `conditions` $(N, 10)$, sequential-count mask, event weights (NLO signs preserved), binary class label. Non-negativity clipping matches EveNet's `log1p` scaling. `--smoke` mode fabricates schema-valid events without `uproot`.
- **Systematic injector** `scripts/inject_systematic.py`: analytic distortions of jet tokens — `jes` (constant scale $\times(1+\alpha)$) and `jes_mhh` ($\alpha \tanh((M_{\text{all}} - m_0)/w)$, a shape systematic correlated with the κ_λ observable) — with globals recomputed. Closure is exact by construction, so P2 value is *measurable* without real variation MC. Known TODO: JES$\rightarrow$MET propagation deliberately omitted in v1.
- **Sweep generator** `scripts/make_klambda_configs.py`: 3 modes (finetune / frozen / scratch) $\times$ 5 dataset fractions $(0.01, 0.03, 0.1, 0.3, 1.0)$ $\times$ 5 seeds = 75 configurations + launcher script; adapted from the EveNet Exotic-Higgs study template.
- **Configs**: `workflow_klambda.yaml` (control file; paths are placeholders) and `event_info_klambda.yaml`, whose INPUTS block is byte-identical to EveNet's `pretrain.yaml` so the 20M backbone binds (assignment/permutation heads present but off).
- **Evaluation** `scripts/plot_data_efficiency.py`: weighted AUC vs. training fraction, mean$\pm$std over seeds, with kinematic-ceiling overlay. *Currently AUC-only* — the closure-metric upgrade (Gate G0.5) is the key missing piece.
- **Runtime**: Perlmutter shifter image `avencast1994/evenet:1.5`; checkpoints via `hf download Avencast/EveNet`.

Engineering TODOs carried from the harness handoff: create `options_frozen.yaml`; verify the prediction key `classification/klambda`; load-test the configs inside the shifter image before launching the 75 jobs; and verify Delphes branch names against real files.

7 Options and costs

On Option C, note also the existence of TauFM [11] (jet-FM-based τ reconstruction) as adjacent prior art, and that the EveNet group invites process contributions and ships a 100M-parameter template — if Option C is ever reached, upstream collaboration dominates a solo rebuild.

	A. Fine-tune stock EveNet (+ corner encoding)	B. Vocabulary surgery + continued pre-training	C. New τ/γ-aware FM
What it is	Existing 75-config harness; optional corner encoding (Sec. 5)	Widen token to 8–9 dims (`isTau`/`isPhoton` or τ-ID score); zero-init new input columns (function-preserving at init); continue SSL+classification pre-training on a τ/γ-enriched Delphes corpus ($Z \to \tau\tau$, $H \to \tau\tau$, γ+jets, $H \to \gamma\gamma$, $t\bar{t}\gamma$, HH grid)	Full pre-training from scratch with τ/γ in vocabulary and corpus
Weight reuse	Full	Full except first linear layer	None
New corpus	None	Modest: F-volume result of [2] (flat low-label scaling) suggests $\mathcal{O}(10^7)$ events from the existing Delphes pipeline plausibly suffices	EveNet-scale was 33B raw / 500M retained; $\mathcal{O}(10^2$–$10^3)\times$ A–B
Compute	Fine-tune only: hours–days on Perlmutter	+ continued pre-training of an 18.8M model: days–weeks	Months + generation campaign (EveNet used 512 A100s)
Key risk	τ/γ information ceiling (bounded by Sec. 4)	No established HEP recipe for continued pre-training; representation drift; schema fork diverges from upstream	Duplicates $\sim$1 year of engineering; **and if A nulls, C likely nulls too** (Sec. 8)

Table 3: The three strategic options.

8 Staged decision plan

Cheapest-information-first. Each gate has an explicit pass condition and a kill/branch interpretation.

G0 — Run the existing sweep (Option A, stock or corner schema).
Pass: finetune > scratch beyond the compute penalty **and** beats the kinematic-ceiling baseline on value-per-compute. *Interpretation of a null:* event-level pre-training did not fix the Sophon null ⇒ the *objective* mismatch is binding (Sec. 2.1); no amount of τ-vocabulary repairs an objective mismatch, so **a null here also condemns Option C in its like-for-like form** — the missing τ-ID is not what limits κ_λ-separation (Sec. 4); the surviving FM directions would be coupling-aware pretexts, not coverage extensions.

G0.5 — Add calibration/closure metrics to the sweep evaluation.
Integral closure < 1% per basis process and shape closure at `CLOSURE_TOL`= 0.05, alongside AUC, for every (mode, fraction, seed) cell. Near-zero marginal cost; fills the literature's largest gap (Sec. 2.2); publishable in either direction; directly feeds the low-statistics-systematics "money plot" goal. *Branch:* if FM fine-tunes win AUC but fail closure where scratch passes, the FM inductive bias is biasing the ratio — a new, reportable failure mode, and a hard blocker for NSBI adoption regardless of G0. *Decision primacy (fixed 2026-07-12, before any sweep curves existed):* the adoption metric is the closure **left-shift** — finetune passing both gates at

a smaller training fraction than scratch (e.g. 3% vs. 30% = a 10× equivalent-data multiplier) *is* an adoption-grade win even at AUC parity; an AUC-only win with failing closure is not. AUC is a secondary guard (finetune must not materially regress below scratch). Calibration, not ordering, is the binding axis: the full-statistics ceiling measurement shows AUC saturating by the 10% fraction (0.792 at 1% vs. 0.817 at 100%) while its closure gates first pass only at 10%, with a method-limited |IC| floor of ≈1.7%.

G1 — Corner-encoding A/B test (τ_h: anonymous vs. $(0, 0, \pm 1)$).
One adapter flag; run at two dataset fractions × 5 seeds. *Pass:* corner ≥ anonymous at all fractions (watch R2 of Sec. 5.3 at the lowest fraction).

G2 — $b\bar{b}\gamma\gamma$ increment.
Photons via $(0, 1, 0)$; only the adapter's object-filling changes. Tests whether the corner trick generalises to a second unseen type, on the channel where the kinematic-preservation argument is strongest.

G3 — Option B surgery (only if corners measurably saturate).
Zero-init input widening + continued SSL pre-training; corpus sized modestly per the flat-volume result. *Pass:* beats the corner encoding by more than the added compute.

G4 — Option C (only if transfer is proven valuable *and* surgery demonstrably fails).
Prefer collaboration with the EveNet group over a solo build.

Standing constraint: no σ_{κ_λ} significance claims from any branch until the upstream NSBI MC-statistics estimator critique [12] is settled (the wifi-ensemble programme [13] is the candidate resolution and is deferred to a separate project).

9 Open questions, ranked

Priority tags [P0/P1/P2] and effort tags [S/M/L] follow the note-family convention.

Q1. **[P0, S] Calibration of fine-tuned FM classifiers.** Does a pre-trained backbone help, hurt, or not affect ratio closure at fixed AUC? Untested in [2] (rejection metrics only) and in [1] (SIC only). Answered by G0.5.

Q2. **[P0, S] Does event-level pre-training beat scratch at all for a continuous interference-driven ratio?** The horizon-vs-objective disentangling test (G0). All published EveNet wins are discrete classification/anomaly tasks.

Q3. **[P0, — (blocking for claims)] MC-statistics estimator critique** [12]: settle before quoting any κ_λ significance from FM or non-FM pipelines.

Q4. **[P1, S] Corner vs. anonymous encoding** (G1), including the low-N_{ft} overfit risk R2.

Q5. **[P1, S] P2 value measurement.** Data-efficiency of nominal-vs-varied ratio learning under the analytic injector: does the FM advantage *grow* as variation-sample size shrinks (the money plot), and does it survive the `jes_mhh` shape systematic that is correlated with the κ_λ observable?

Q6. **[P1, M] Quantify the FastMTT/$m_{\tau\tau}$ and τ-ID loss.** Compare the kinematic ceiling with and without $m_{\tau\tau}$/ID features to bound what the stock schema forfeits in absolute terms; separately, test whether EveNet's neutrino-reconstruction pretext implicitly recovers $m_{\tau\tau}$-like information (probe the fine-tuned representation).

Q7. **[P1, S] Parameterised-head caution.** EveNet's mass-parameterised fine-tuning *reduced* sensitivity; verify our per-basis-point + linear-combination NSBI construction is unaffected, and treat any κ_λ-parameterised head proposal as requiring its own gate.

Q8. **[P2, M] Continued-pre-training recipe stability** (for G3): zero-init widening, replay mixture of original-corpus processes vs. τ/γ additions, representation-drift monitoring. No established HEP precedent.

Q9. **[P2, S] Domain gap: generic Delphes card vs. `cms_card_v1`.** Empirically bridged in the paper (generic→CMS full-sim); verify on our samples, since our card is Run-3-tuned (UParT/DeepTau baselines).

Q10. **[P2, M] Value-per-compute accounting across the NSBI multiplicity** (processes × systematics × ensembles × κ_λ grid), including whether a *single* fine-tuned backbone can be shared across ratio components (amortisation would strongly favour the FM).

10 Summary verdict

- **Coverage:** EveNet does *not* cover τ or γ final states — no vocabulary slot, no corpus processes, no supported extension path. This is established fact, not conjecture (Sec. 2.3).
- **Feasibility anyway:** the gap is concentrated in identification information that approximately cancels in the low-statistics ratio components (signal–signal and nominal–varied) where FM data-efficiency is actually needed; the κ_λ-carrying kinematics survive the stock schema exactly, up to the visible-kinematics caveat ($m_{\tau\tau}$ loss, partially mitigated by the neutrino-reconstruction pretext) (Sec. 4).
- **Do not build a new FM now, and do not start with surgery.** The existing harness is the correct next instrument: it runs the disentangling experiment (G0) whose null would condemn a like-for-like Option C anyway, and — upgraded with closure metrics (G0.5) — it resolves the literature's largest untested assumption for FM-based NSBI at near-zero marginal cost.
- **Cheap upgrades first:** the unoccupied-corner encoding (τ_h as $(0, 0, \pm1)$, γ as $(0, 1, 0)$) recovers a type signal with zero weight surgery and one adapter line; it strictly dominates the anonymous-jet encoding at the data level, with finite-sample risks that G1 measures directly.
- **Escalation is evidence-gated:** corners → zero-init widening + modest continued pre-training (volume de-risked by the flat low-label scaling result) → new/collaborative FM, each step only on demonstrated saturation of the previous one.

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